Warehouse Allocation and space analysis

Background: Zenith Logistics Pvt. Ltd. is a mid-sized 3PL (Third Party Logistics) provider headquartered in Gurugram, India. Established in 2008, the company specializes in contract logistics, warehousing, and last-mile distribution. With strong clients in FMCG, e-commerce, and pharmaceuticals, Zenith has grown rapidly, managing over 200,000 sq. ft. of warehouse space across North India. Its flagship facility in Delhi NCR (10,000 sq. ft.) is strategically located near NH-48 and the IGI Airport, enabling both regional and national distribution. This warehouse is considered a *"high-revenue site"* because of its proximity to key consumer markets like Delhi, Gurugram, Noida, and Ghaziabad.

Business Environment: The warehousing industry in India is undergoing transformation due to GST-driven consolidation, e-commerce growth, and rising real estate costs. Clients are demanding flexible contracts and higher service levels. Zenith competes with large players like DHL, Delhivery, and Mahindra Logistics. To stay competitive, it emphasizes customized contracts and flexible space allocation.

Cost Structure: Fixed monthly cost of operating the NCR warehouse: ₹20 lakhs (rent, utilities, manpower).

Break-even utilization level: 85% (8,500 sq. ft.).

Any underutilization below this level directly impacts profitability.

The Manager's Challenge: The Warehouse Operations Manager, Ankit Verma, has to allocate 10,000 sq. ft. of space among 10 diverse clients (A–J), as given in Table below.

Client	Need (sqft)	Rate (₹/sqft)
A-pharma	1200	14
B-ecom	800	9
C-pharma	1500	11
D-pharma	2000	13
E-FMCG	900	8
F-pharma	1300	12
G-ecom	1100	10
H-FMCG	1700	7
I-pharma	1400	15
J-startup	1800	6

Some are premium clients (e.g., Pharma distributors) willing to pay up to ₹11-15/sq. ft. Others are volume clients (e.g., FMCG, e-commerce startups) who demand large areas but pay as low as ₹6–₹8/sq. ft. Company policy requires at least 50% fulfillment of each client's request to maintain long-term

partnerships. Each client gets at least 50% of requested needs. Minimum utilization floor: 90% (9,000 sq. ft.) to cover fixed costs. Mr. Ankit wants to maximize monthly rental revenue, while maintaining strategic client relationships and high utilization.

Compute: Most optimal space allocation to improve warehouse space utilization and fulfilling customer demand.